

Synopsis Document

Project Members

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1. Introduction to the Project

Human Resource Management (HRM) is one of the most important functions in every organization, responsible for managing employee information, recruitment, attendance, leave, payroll, performance, client relationships, and organizational operations. As organizations expand, handling these activities through multiple independent applications or manual processes becomes inefficient, leading to data duplication, increased administrative effort, poor collaboration, and higher operational costs.

Lexorion Horizon is an Enterprise Human Resource Management System (HRMS) developed using a Microservices Architecture. The platform integrates recruitment, workforce management, finance, client management, notifications, reporting, and organizational administration into a unified ecosystem while keeping each business capability as an independent microservice.

Unlike traditional monolithic applications, where all functionalities are tightly coupled, Lexorion Horizon separates each business domain into dedicated services such as Authentication, Organization, Recruitment, Workforce, Finance, Client, Notification, and Reporting. Each service maintains its own database and communicates through REST APIs, allowing independent development, deployment, maintenance, and scaling.

The system also follows a Multi-Tenant Architecture, enabling multiple organizations to securely use the application while ensuring complete isolation of organizational data. This approach provides better scalability, improved fault isolation, simplified maintenance, and greater flexibility for future expansion.

Lexorion Horizon is designed for staffing agencies, recruitment firms, consulting organizations, service-based companies, and small to medium-sized enterprises seeking a modern, secure, and scalable HRMS.

2. Problem Statement

Many organizations continue to rely on separate software applications for recruitment, employee management, payroll, attendance, finance, and reporting. In some cases, spreadsheets and manual paperwork are still used to manage daily HR activities. Such fragmented systems create operational inefficiencies and make organizational growth difficult.

The major challenges include:

- Duplicate and inconsistent employee records.
- Manual data entry across multiple applications.
- Poor integration between HR, recruitment, finance, and administration.
- Limited scalability of monolithic applications.
- Difficult software maintenance and upgrades.
- Lack of centralized reporting and analytics.
- Security risks caused by inconsistent user access control.
- Increased operational costs and reduced productivity.

Lexorion Horizon addresses these challenges by providing a centralized Enterprise HRMS based on Microservices Architecture. Each business function is implemented as an independent service, allowing organizations to scale modules independently, simplify maintenance, improve security, and automate HR workflows while maintaining data consistency across the platform.

3. Objectives

The primary objective of Lexorion Horizon is to develop a scalable, modular, and secure Enterprise Human Resource Management System using Java and Spring Boot with a Microservices Architecture.

The specific objectives are:

- Develop an Enterprise HRMS using Java and Spring Boot.
 - Implement independently deployable microservices.
 - Support multiple organizations through Multi-Tenant Architecture.
 - Automate recruitment from job requirements to employee onboarding.
 - Manage employee attendance, leave, performance, assets, and payroll.
 - Manage client relationships, contracts, invoices, and payments.
 - Provide secure authentication and role-based authorization.
 - Improve scalability through Microservices Architecture.
 - Reduce manual administrative work using integrated workflows.
 - Generate centralized business reports.
 - Build a flexible platform capable of future expansion.
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4. Scope of the Project

Lexorion Horizon provides a complete Human Resource Management ecosystem by integrating multiple business services into one platform.

Authentication & Authorization

- User Registration
- User Login
- Role Management
- Permission Management
- Session Management

Organization Management

- Organization Registration
- Branch Management
- Department Management
- Team Management
- Organization Settings

Recruitment Management

- Job Requirement Management
- Candidate Registration
- Application Management
- Interview Scheduling
- Offer Management
- Candidate Placement

Workforce Management

- Employee Information Management
- Attendance Management
- Leave Management
- Performance Management
- Asset Management

- Payroll Management
- Employee Lifecycle Management

Finance Management

- Client Billing
- Invoice Management
- Payment Processing
- Expense Management
- Financial Reporting

Client Management

- Client Registration
- Client Contact Management
- Client Contract Management
- Requirement Management

Notification Management

- Email Notifications
- SMS Notifications
- System Alerts
- Workflow Notifications

Reporting

- Recruitment Reports
- Employee Reports
- Attendance Reports
- Payroll Reports
- Financial Reports
- Operational Reports

Project Limitations

The following features are outside the scope of the current phase and may be implemented in future versions:

- Mobile Applications
 - Third-Party Integrations
 - Cloud Deployment
 - Advanced Workflow Automation
 - AI-based Resume Screening
 - Industry-Specific Extensions
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5. Technologies Used

Lexorion Horizon is developed using modern enterprise technologies and open-source frameworks that support scalable application development and Microservices Architecture.

Layer	Technology
Frontend	React.js
Programming Language (Frontend)	TypeScript
UI Framework	Tailwind CSS
Icons	Lucide React
Backend Language	Java 21
Backend Framework	Spring Boot 3.x
Microservices Framework	Spring Cloud
API Gateway	Spring Cloud Gateway
Service Discovery	Eureka Server
Security	Spring Security
Authentication	JWT (JSON Web Token)
Database	PostgreSQL
ORM Framework	Spring Data JPA (Hibernate)
Cache	Redis
Message Broker	Apache Kafka
API Documentation	Swagger / OpenAPI
Build Tool	Maven
Version Control	Git
Source Code Repository	GitHub
IDE	IntelliJ IDEA Community Edition / Visual Studio Code
Database Management	pgAdmin 4
API Testing	Postman
UI Design	Figma
Operating System	macOS

The selected technology stack provides modularity, security, scalability, maintainability, and efficient development while using reliable open-source tools suitable for enterprise software.

6. Expected Outcome

The successful implementation of Lexorion Horizon will provide organizations with a centralized, secure, and modular Human Resource Management System capable of managing the complete employee lifecycle.

The system is expected to:

- Centralize recruitment, workforce, finance, and client operations.
- Improve recruitment efficiency and employee lifecycle management.
- Reduce manual administrative work through workflow automation.
- Simplify attendance, leave, payroll, and finance management.
- Provide secure role-based access control.
- Support multiple organizations through Multi-Tenant Architecture.
- Enable independent deployment and scaling of individual microservices.
- Reduce maintenance complexity using modular application design.
- Provide centralized reporting for improved business decision-making.
- Establish a flexible foundation for future enterprise enhancements.

Upon completion, Lexorion Horizon will demonstrate how Microservices Architecture can be effectively used to develop a scalable, maintainable, secure, and enterprise-ready Human Resource Management System suitable for modern organizations.